Robust Trajectory Tracking for Autonomous Skid-Steer Mobile Robots

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Overview

1. Problem Formulation
2. System Architecture
3. Contact Force Modeling
4. Control Architecture
   - Lyapunov + SMC + STA
   - Backstepping + SMC + STA
   - NMPC + ADRC (LESO)
5. System Implementation
6. Perfomance Evaluation
   - Lyupunov Performance
   - Backstepping Performance
   - NMPC+ADRC+LESO Performance
   - Tracking Robustness on Discontinuous Trajectories
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7. Future Work
8. Bibliography

Problem Formulation

Motivation:

- Skid-steer robots are widely used in rough terrain and mobile robotics.
- Simple mechanical design but **complex interaction with ground**.

Lateral Slip Effects on Stability:

- Turning requires differential wheel speeds which induces **skidding**.
- No pure rolling condition during turns.
- Significant **lateral velocity** not directly controllable.
- Mismatch between commanded and actual motion causes instability, oscillations, tracking errors.

System Architecture: URDF & Parameters

URDF Model

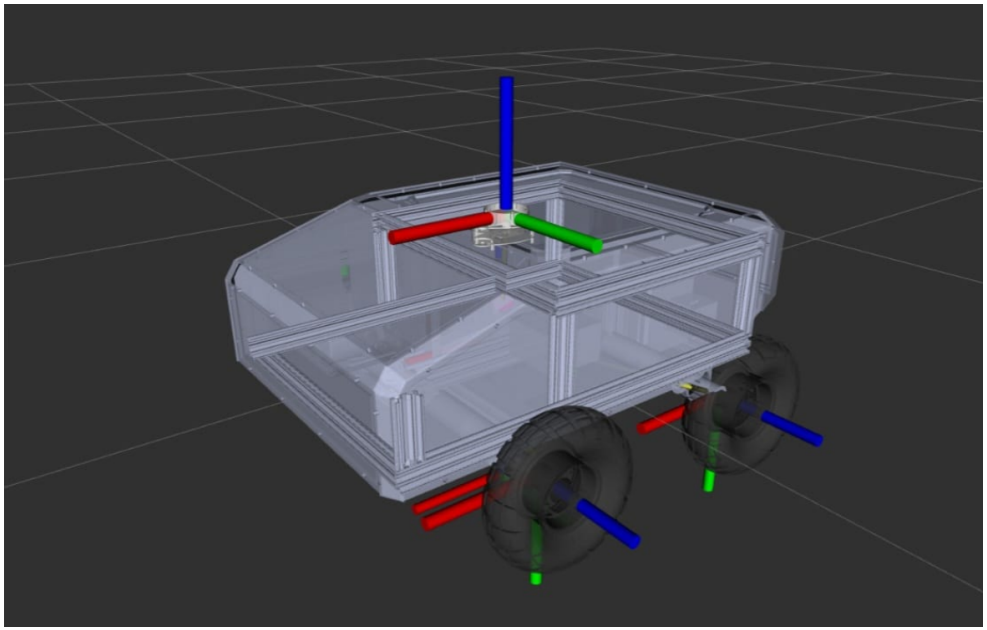


Fig. 1: URDF model of the skid-steer mobile robot

System Parameters

Geometric Parameters

Parameter	Symbol	Value
Wheel radius	r	0.10 m
Half track width	b	0.3055 m
Lookahead distance	d	0.35 m
Wheelbase	L	0.420 m

Dynamic Parameters

Parameter	Symbol	Value
Mass	m	38.805 kg
Moment of inertia	I_z	0.751 kg $\cdot$ m^2
Center of mass	(x_c, y_c, z_c)	$(-0.075, -0.029, 0.077)$ m

Contact Force Modeling (Simscape Multibody)

Ground Interaction Model

- Modeled using Simscape **Spatial Contact Force** blocks
- Target environment: harsh roads (wet asphalt / loose gravel)
- Penalty-based *smooth stick–slip* friction model.

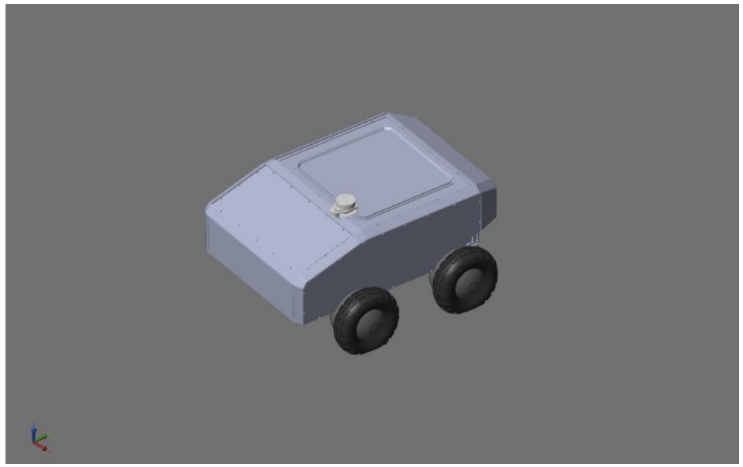


Fig. 2: Multibody 3D visualization

Simscape Parameters [1]

Parameter	Value
Static friction (μ_s)	0.5
Dynamic friction (μ_k)	0.4
Contact stiffness	1×10^5 N/m
Damping	5 N/(m/s)

1. WMR Kinematic Model

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -d\sin\theta \\ \sin\theta & d\cos\theta \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \nu_o \\ \omega_o \end{bmatrix}$$

where d denotes the center-of-mass offset

2. Error Kinematics

Tracking error w.r.t. reference $(\nu_r, \omega_r, \theta_r)$

$$\dot{x}_e = y_e\omega_o + \nu_r\cos\theta_e - d\omega_r\sin\theta_e - \nu_o$$
$$\dot{y}_e = -x_e\omega_o + \nu_r\sin\theta_e + d\omega_r\cos\theta_e - d\omega_o$$
$$\dot{\theta}_e = \omega_r - \omega_o$$

3. Skid-Steer Robot Dynamics

$$M(p(t))\ddot{p}(t) + C(p(t),\dot{p}(t))\dot{p}(t) + G(p(t)) + \tau_d(t) = B(p(t))\tau(t) - A(p(t))\lambda$$

- $p = [x, y, \theta]^T$: robot pose
- $M(p)$: inertia matrix
- $C(p, \dot{p})$: Coriolis/centripetal effects
- $G(p)$: gravity (0 for planar motion)
- $\tau = [\tau_l, \tau_r]^T$: wheel torques
- τ_d: external disturbance

1. Kinematic Level: Lyapunov Control

Guarantees asymptotic stability for nonholonomic WMR errors (x_e, y_e, θ_e).

$$\nu_c = k_1(x_e + d(1 - \cos\theta_e)) + \nu_r \cos\theta_e$$
$$\omega_c = \omega_r + \Xi + k_3 \sin(\theta_e/2)$$

$$\Xi = k_2(y_e - d\sin\theta_e)\,\nu_r \cos(\theta_e/2)$$

Signal Filtering: LTD

Linear Tracking Differentiator (LTD) acts as a low-pass filter to extract smooth derivatives $(\dot{v}_d, \ddot{v}_d)$.

2. Dynamic Level: LESO + STA-SMC

Disturbance Observer (LESO)

$$\dot{z}_1 = z_2 + \beta_1(v - z_1)$$
$$\dot{z}_2 = z_3 + b_0 u + \beta_2(v - z_1)$$
$$\dot{z}_3 = \beta_3(v - z_1)$$

z_3: lumped disturbance estimate

Super-Twisting SMC

$$u_{eq} = \ddot{v}_d - \lambda\dot{\tilde{v}} - z_3$$
$$u_{sw} = -k_1\sqrt{|s|}\,\mathrm{sign}(s) - \int k_2 \mathrm{sign}(s)\,dt$$

$$u = \delta^{-1}(u_{eq} + u_{sw})$$

1. Kinematic Level: Adaptive Backstepping

Generates bounded velocity commands for posture error (x_e, y_e, θ_e). [2]

$$v_c = v_r \cos(\theta_e) + k_1 x_e$$
$$\omega_c = \omega_r + k_2(|v_r| + \epsilon) y_e + k_3(|v_r| + \epsilon) \sin\theta_e$$

Adaptive ICR shift [3]:

$$d(t) = X_{icr_max} \exp(-\alpha|\kappa| v_r^2), \quad \kappa = \frac{\omega_r}{v_r}$$

Wheel velocity mapping:

$$v_{ld} = v_c - d(t)\omega_c - b\,\omega_c$$
$$v_{rd} = v_c - d(t)\omega_c + b\,\omega_c$$

2. Dynamic Level: LESO + STA-SMC

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Super-Twisting SMC

$$u_{eq} = \ddot{v}_d - \lambda\dot{\tilde{v}} - z_3$$
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$$u = \delta^{-1}(u_{eq} + u_{sw})$$

III. NMPC: Kinematic Layer(CasADi + IPOPT)

Error (Body Frame)

$$x_e = c_\theta(x_r - x) + s_\theta(y_r - y)$$
$$y_e = -s_\theta(x_r - x) + c_\theta(y_r - y)$$
$$\theta_e = \theta_r - \theta$$

Error Dynamics

$$\dot{x}_e = \omega y_e - v + v_r c_{\theta_e}$$
$$\dot{y}_e = -\omega x_e + v_r s_{\theta_e}$$
$$\dot{\theta}_e = \omega_r - \omega$$

Optimization

$$\boxed{J = \sum_{k=1}^{N} \left(x_{e,k}^T Q x_{e,k} + u_k^T R u_k \right)}$$

$$Q = \mathrm{diag}(200, 200, 5), \quad R = \mathrm{diag}(0.1, 0.05)$$

Constraints & Horizon

$$-0.3 \le v \le 0.6, \; -0.7 \le \omega \le 0.7$$
$$N = 20, \quad \Delta t = 0.2\,s$$

III. ADRC + LESO: Dynamic Layer

System Model

$$\boxed{\dot{\omega} = f(t) + b_0 u}$$

$f(t)$: lumped disturbance (friction, terrain, uncertainties)

LESO (State & Disturbance Estimation)

$$\dot{z}_1 = z_2 - \beta_1(z_1 - \omega) + b_0 u$$
$$\dot{z}_2 = -\beta_2(z_1 - \omega)$$

z_1: velocity estimate z_2: disturbance estimate

Tracking Error

$$e = \omega_{ref} - \omega$$

ADRC Control Law

$$u = \frac{1}{b_0}\left(K_p e - z_2\right)$$

Parameters

$$\beta_1 = 100,\ \beta_2 = 400, \quad K_p = 50,\ b_0 = 10.5$$

Robust velocity tracking with real-time disturbance rejection

System Implementation: Control Architecture

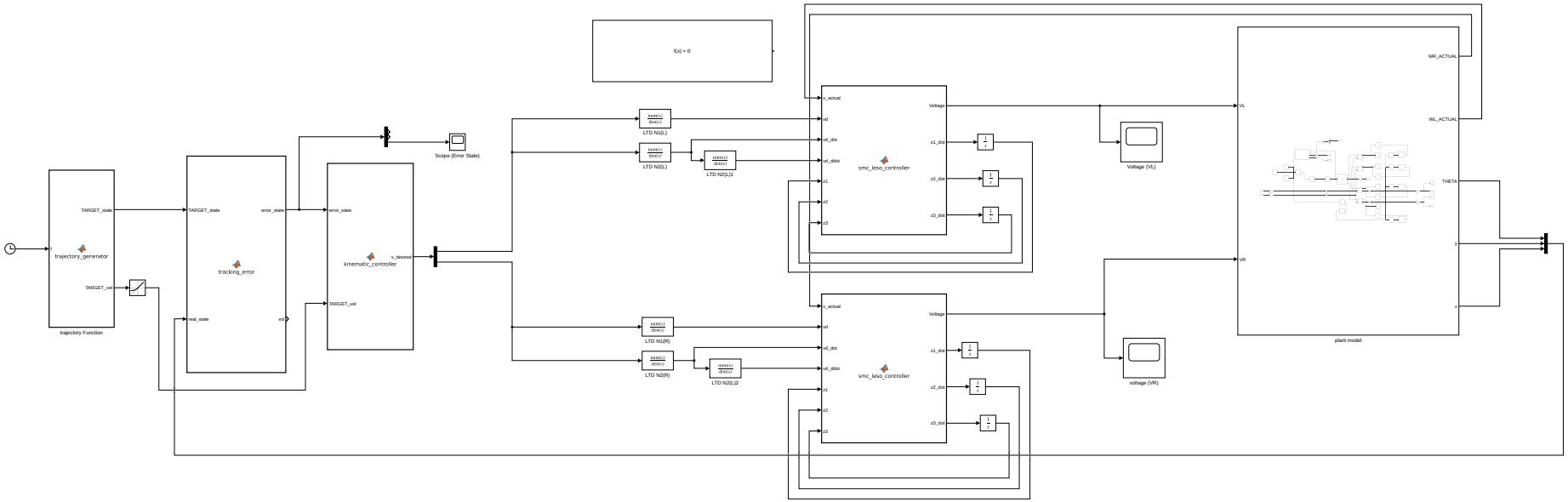


Fig. 3: *Overall Simulink cascaded control architecture*

System Implementation: Physical Plant

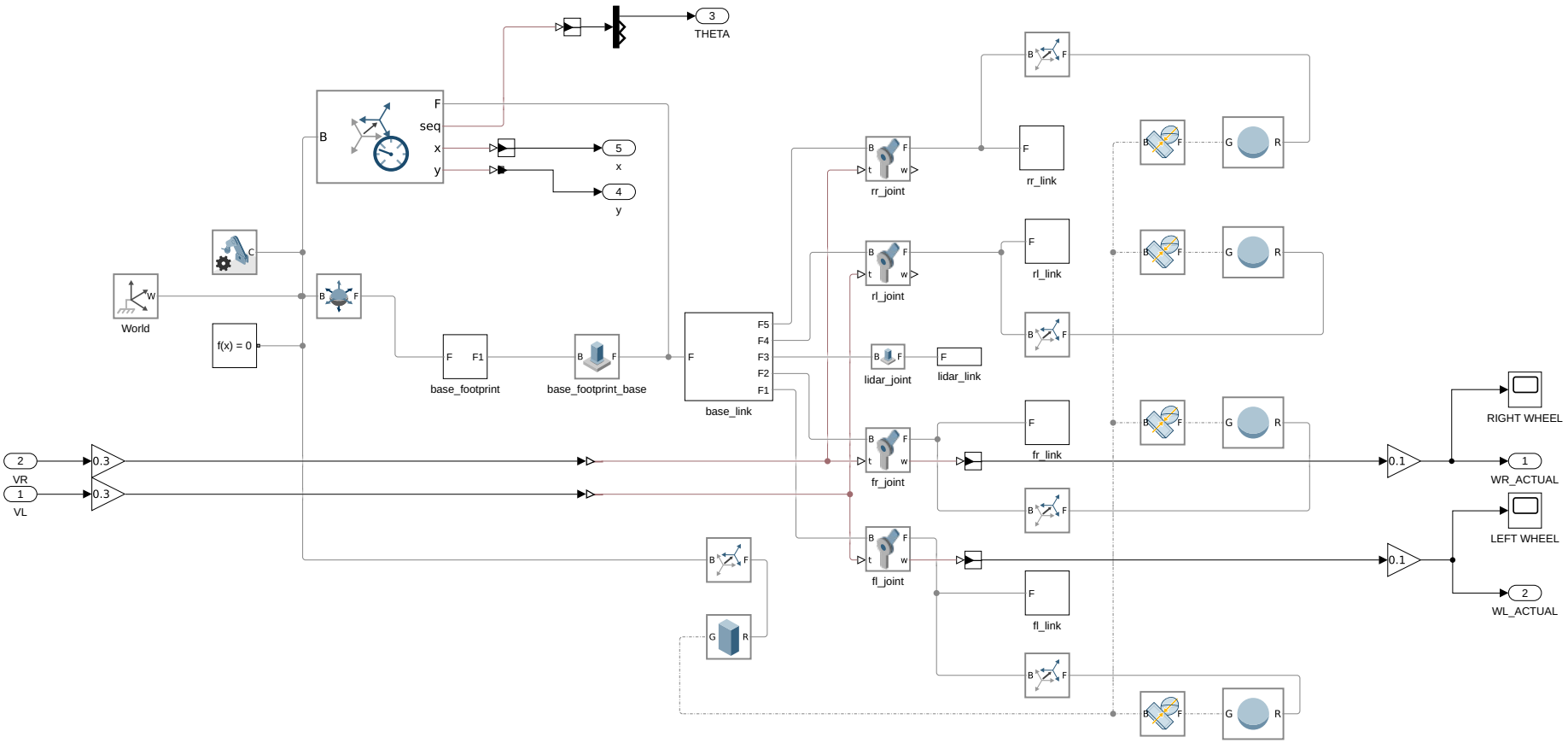


Fig. 4: *High-fidelity Simscape Multibody plant model*

System Implementation: NMPC Control Architecture

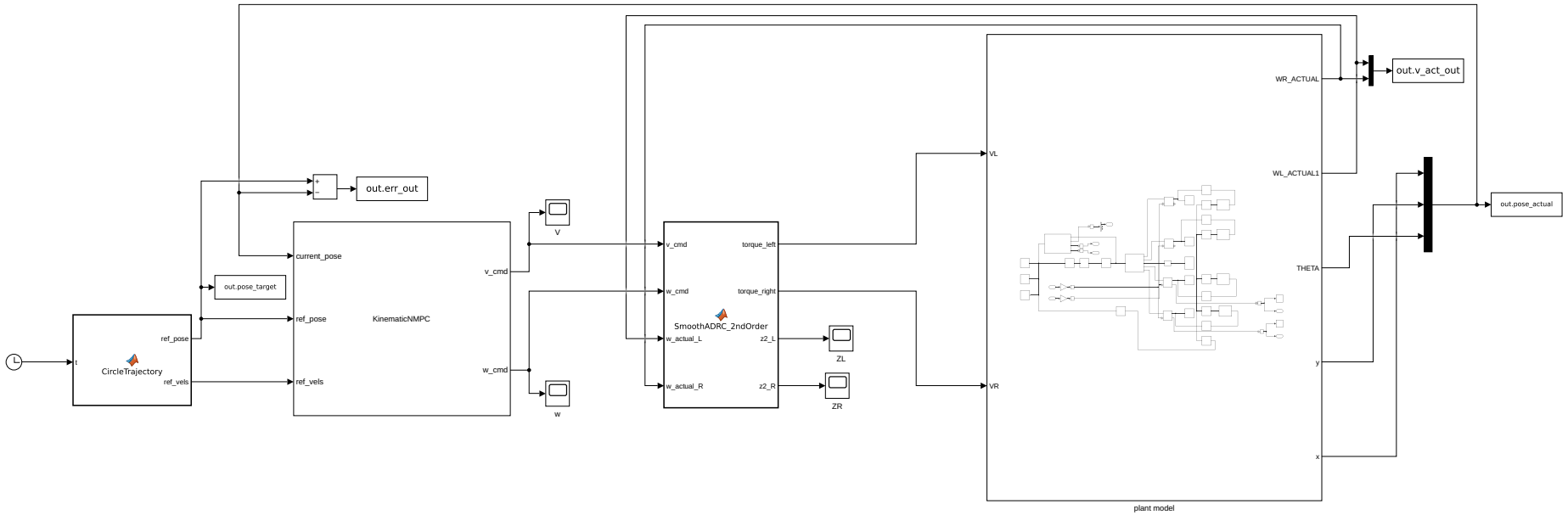
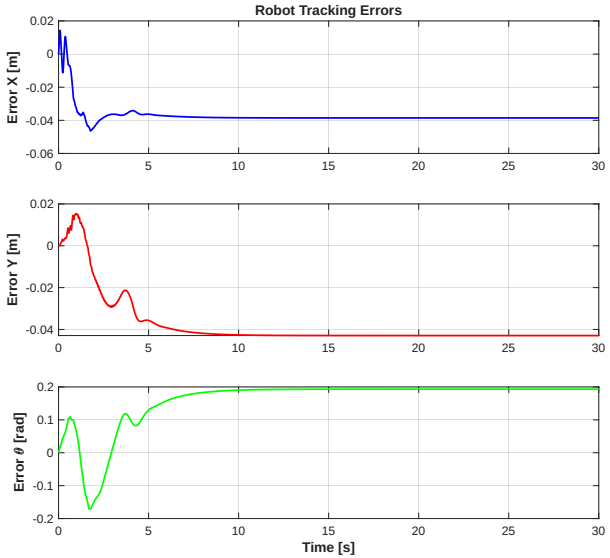
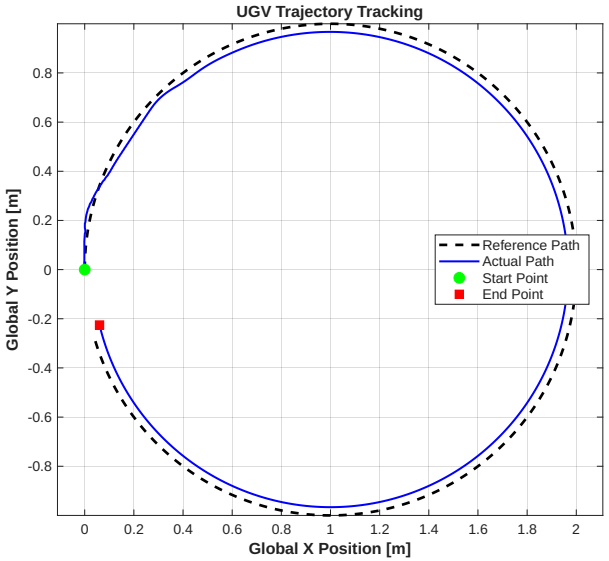


Fig. 5: *NMPC+ADRC+LESO Cascaded Simulink Architecture I*

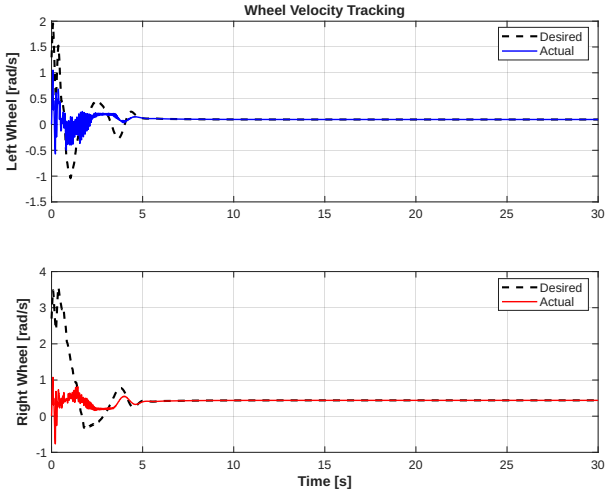
Performance Evaluation :Lyupunov



Tracking Error



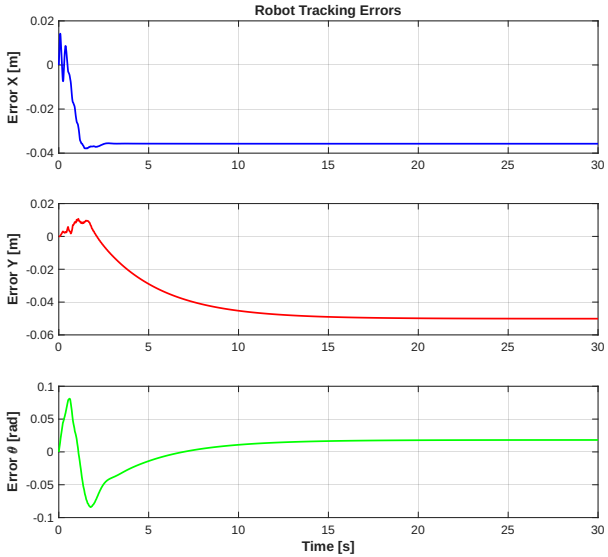
Trajectory Tracking



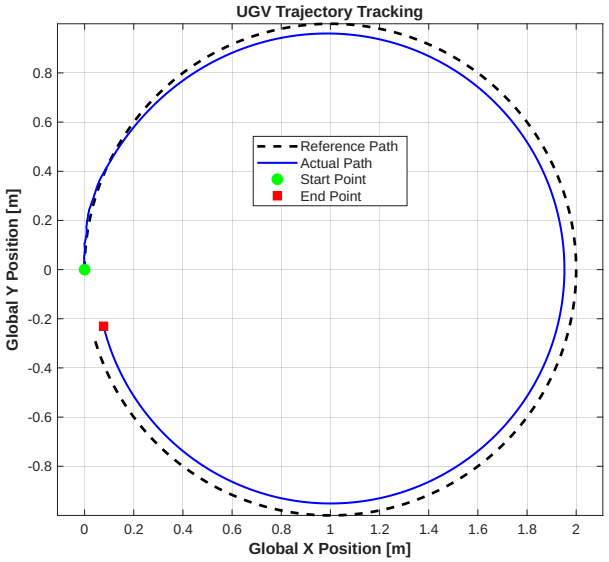
Wheel Velocity

Reference Path: Circular trajectory ($R = 1$ m)

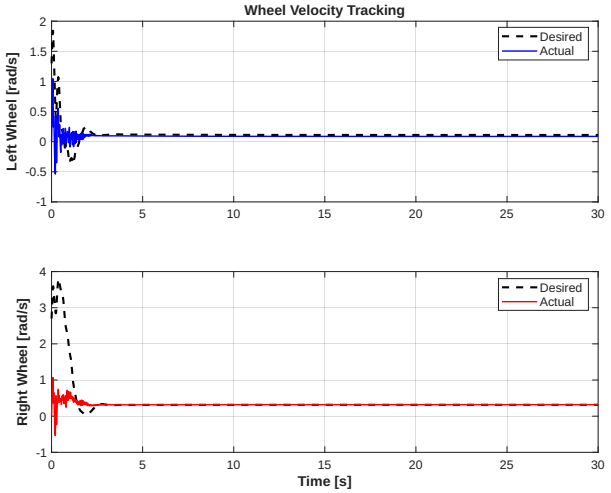
Performance Evaluation :Backstepping



Tracking Error



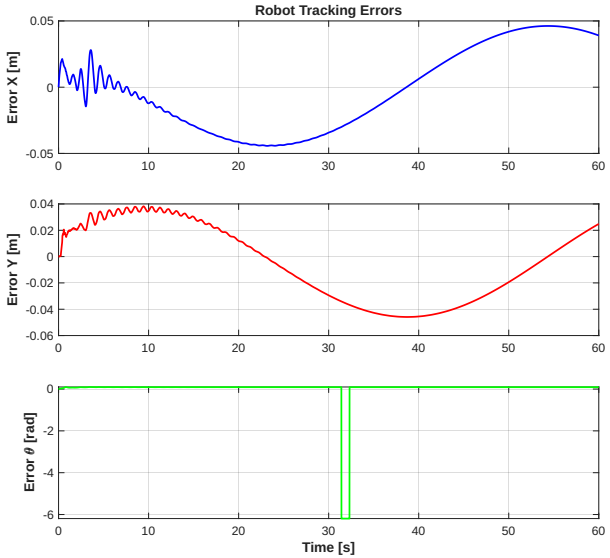
Trajectory Tracking



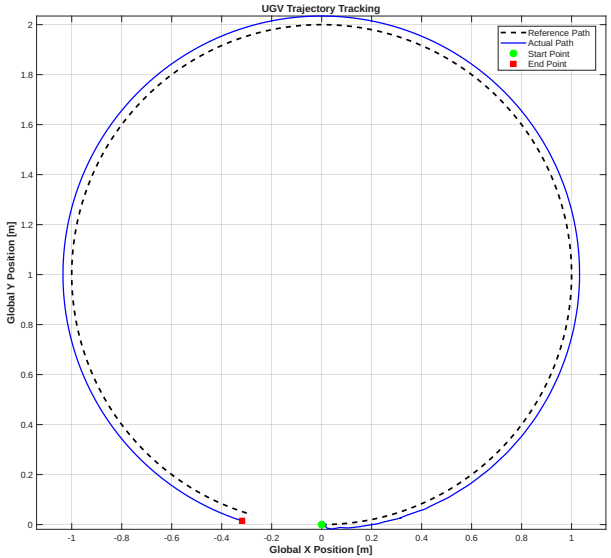
Wheel Velocity

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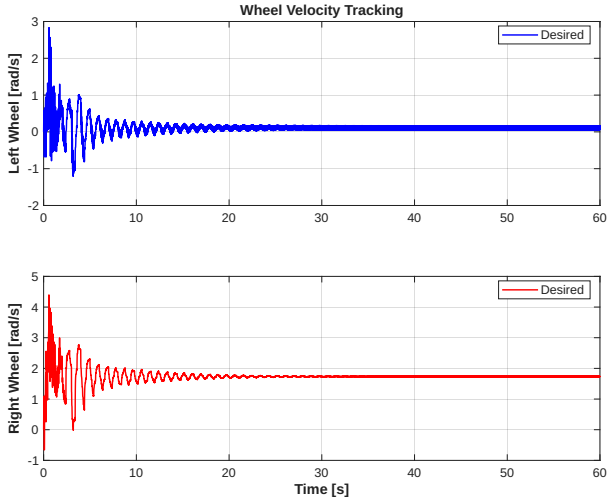
Performance Evaluation :NMPC+ADRC+LESO



Tracking Error



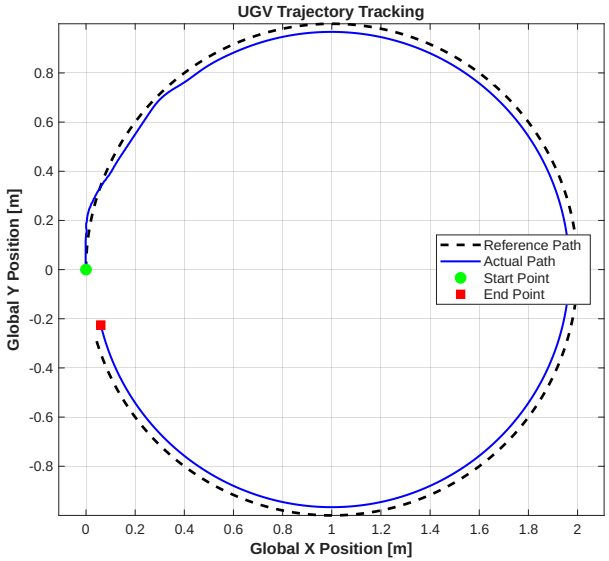
Trajectory Tracking



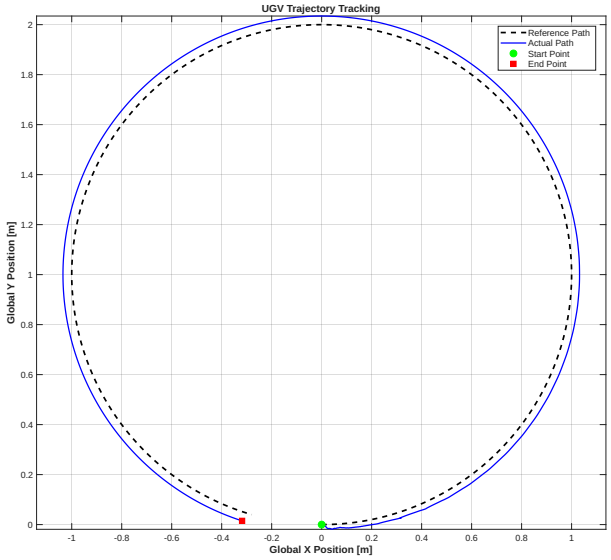
Wheel Velocity

Reference Path: Circular trajectory ($R = 1$ m)

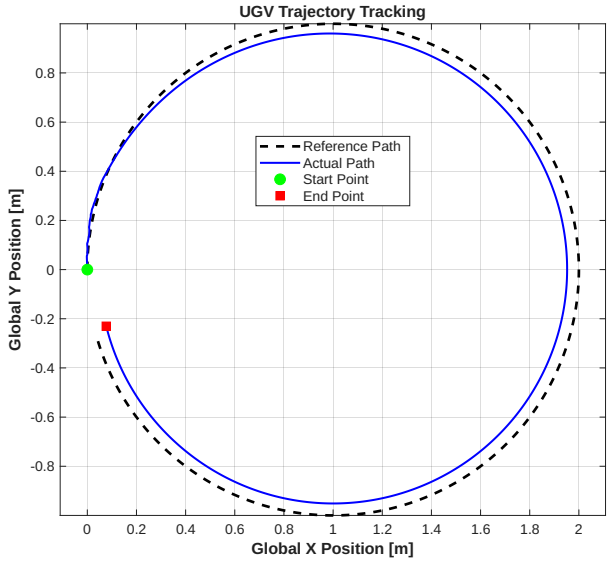
Tracking Robustness on Discontinuous Trajectories



Lyupunov+SMC



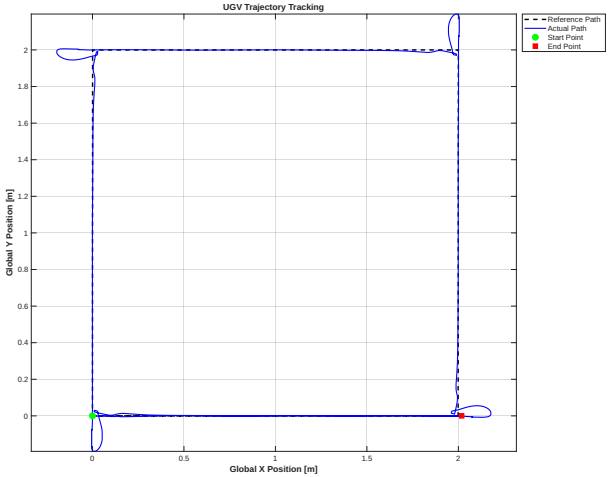
NMPC+ADRC



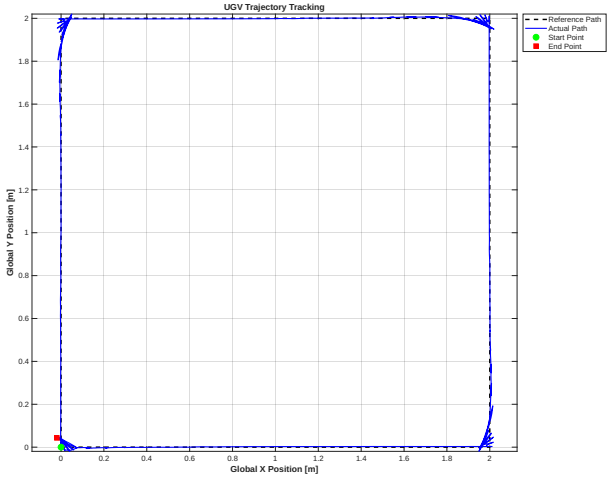
Backstepping+smc

Reference Path: Circular trajectory ($R = 1$ m)

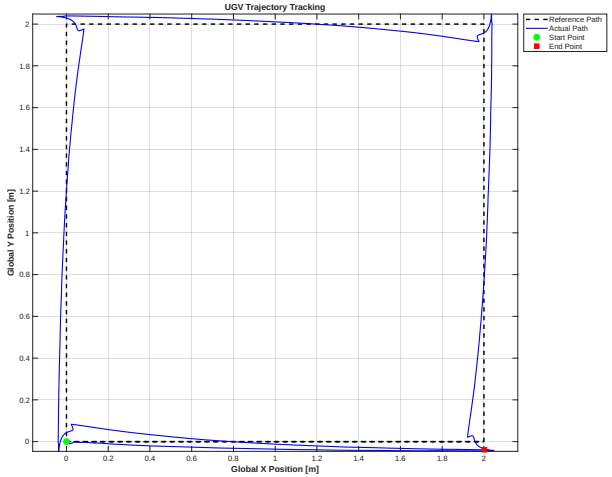
Tracking Robustness on Discontinuous Trajectories



Lyupunov+SMC



NMPC+ADRC



Backstepping+smc

The simulation results clearly demonstrate that the NMPC + ADRC + LESO architecture provides a significant performance leap over the benchmark methods.

By mastering both the smoothness of the circle and the discontinuity of the square, the NMPC+ADRC controller is both predictive (planning the path) and robust (rejecting disturbances).

Future Work

System Deployment & Validation

- **C++ Implementation** of controllers for real-time execution
- Integration with **ROS2 framework** (modular node architecture)
- Deployment and testing in **Gazebo simulation environment**

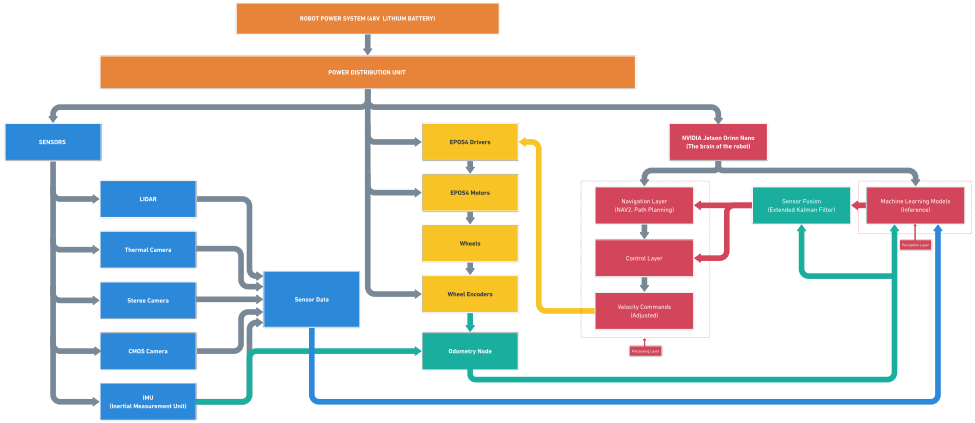


Fig. 6: Proposed ROS2-based system architecture

Research Extensions

- Robustness evaluation under **real-world disturbances**
- Multi-agent coordination for **swarm scenarios**
- Integration of **learning-based perception modules**



Fig. 7: Assembled robotic platform for deployment

References I


K. Popp and W. Schiehlen, *Ground Vehicle Dynamics*. Springer, 2010.


R. Fierro and F. L. Lewis, "Control of a nonholonomic mobile robot: backstepping kinematics into dynamics," *Journal of Robotic Systems*, vol. 14, no. 3, pp. 149–163, 1997.


J.-Y. Jun, M. D. Hua, and F. Ben Amar, "A trajectory tracking control design for a skid-steering mobile robot by adapting its desired instantaneous center of rotation," in *2014 IEEE 53rd Annual Conference on Decision and Control (CDC)*. IEEE, 2014, pp. 4554–4559.